PIYUSH SHAH

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Instrumentation & Control Engineering Technology student with experience in PID tuning, PLC programming, commissioning, sensor calibration, maintenance, and troubleshooting.

WORK EXPERIENCE

BAS Controls Specialist Co-op May 2026 – Sep 2026
Integrated Control Systems Inc. *Winnipeg, Manitoba*

- Programmed closed-loop control sequences (PID loops, setpoint scheduling, interlocks) from Engineer's Sequence of Operation narratives, translating process intent into DDC logic across 4+ commercial projects
- Commissioned 150+ controllers governing AHUs, motors, pumps, and boilers on live BACnet MS/TP and Modbus RTU/TCP networks, verifying each against its control narrative
- Calibrated analog field sensors and transmitters during commissioning
- Troubleshot field faults end-to-end, isolating whether a problem is in the control logic, the network, or the wiring
- Produced 4+ control drawings covering control wiring, field device installation, panel layout, and termination details

Electrical Technician June 2025 – Sep 2025
Tamarack Industries *Winnipeg, Manitoba*

- Wired 100+ control panels for mobile glycol heating units, working from schematic drawings to meet client specs
- Wired 20+ tractors from bare frame to running, including battery and ignition circuits, starter, engine wiring, lighting, horn, and operator controls
- Collaborated with engineers to troubleshoot production issues

TECHNICAL SKILLS

PLCs & Control Platforms: Allen-Bradley CompactLogix, Siemens S7, DeltaV, Arduino
Instrumentation: 4–20 mA signal loops, HART Communicator, sensor calibration, loop commissioning, P&ID interpretation
Protocols & Networking: BACnet MS/TP, Modbus RTU/TCP, EtherNet/IP, OPC-UA, RS-485
Programming: Ladder Logic, Function Block, C++, MATLAB, Python
Software & Tools: AutoCAD, Visio, SolidWorks, LabVIEW, Multisim, MS Office, Grafana, WebCTRL

EDUCATION

Instrumentation and Control Engineering Technology Fall 2024 – Dec 2026
Red River College Polytechnic *Winnipeg, Manitoba*

Bachelor of Computer Science (Completed 2 years) Fall 2022 – Spring 2024
University of Manitoba *Winnipeg, Manitoba*

TECHNICAL PROJECTS

Recipe-Driven Aluminum Casting Control System Aug 2026 – Present

- Developed PLC control software for a four-way permanent mold casting table on a Rockwell Micro850
- Built a counter-based state machine driving the full casting cycle, with HMI recipes configuring motion for two-, three-, and four-cheek molds
- Programmed ultrasonic position feedback with tolerance deadbands, scaled sensor signals to engineering units, and wrote reusable UDFBs for individual and paired cheek motion
- Implemented alarm and diagnostic logic and a layered command path isolating field outputs for safe interlocking

Cascaded Level Control with FactoryTalk HMI March 2026

- Designed a cascade PIDE control strategy on Allen-Bradley CompactLogix (Studio 5000), with an outer tank-level loop driving an inner inlet-flow loop
- Configured PIDE instructions with anti-windup and bumpless auto/manual transfer, then autotuned both loops in sequence and validated against step tests
- Built a FactoryTalk View Studio SE operator interface with live PV/SP/output trends, setpoint entry, and a tiered ISA-18.2 style alarm banner

Closed-Loop PID Temperature Control (IMC Tuning) April 2026

- Identified an FOPDT process model from step-response tests on an Arduino heater and thermistor setup
- Calculated initial PID gains using IMC tuning, then refined them through iteration
- Achieved ~2% overshoot with no steady-state error and no sustained oscillation